

Traceable, but Not Sustainable? Why Digital Visibility Fails in Fragmented Supply Chains

Keywords. Digital traceability; circular economy; PLS-SEM; agro-manufacturing; transparency versus abatement

Abstract

Net-zero manufacturing relies on a calculative infrastructure that most production networks lack. Conventional accounting treats unrecovered material losses as non-existent whenever the observing actor cannot recover them. This study operationalises inter-organisational circular accounting capability, separates it from digital traceability, and evaluates how information turns into action using PLS-SEM on survey data from 520 agro-manufacturing enterprises with 10,000 bootstrap resamples. Calculation drives visibility ($\beta = 0.532$) about three times more strongly than digital traceability, and coordination creates value ($\beta = 0.509$) with traceability amplifying that path ($\beta = 0.325$ direct, $\beta = 0.074$ interaction). Visibility alone yields no action ($\beta = -0.052$, $p = 0.145$), whereas governance directly enables coordination ($\beta = 0.516$). Transparency therefore does not automatically trigger resource recovery.

1. Introduction

Net-zero manufacturing depends on a calculative infrastructure that most production networks do not have. Conventional accounting records what a firm buys and what it disposes of, and it stops at the legal boundary of the entity. Material that crosses that boundary and cannot be recovered by the observing actor is never recorded. The loss stays invisible, not because it is small, but because no single actor holds both the capacity to recover it and the mandate to measure it.

Policy has answered this silence with disclosure. Digital product passports, deforestation-free sourcing rules and sustainability reporting standards all rest on one causal chain. Better data makes flows visible, visibility exposes losses, and exposed losses get recovered. The chain is intuitive and rarely tested. Firms keep investing in tracking technology while recovery rates stay flat, which suggests the chain breaks between seeing and acting.

This study tests where it breaks. Inter-organisational circular accounting capability is defined as the shared ability of buyers and suppliers to measure, allocate and value material losses across ownership boundaries. Digital traceability is defined separately as the technical ability to identify physical units and follow their movement. The two are usually treated as one thing, and separating them allows a direct comparison.

Three questions follow. First, what makes material flows visible. Second, whether visibility alone produces recovery action. Third, what converts coordination into circular value. Evidence comes from 520 agro-manufacturing enterprises, a setting of fragmented ownership, seasonal supply and high loss rates, where the measurement problem is severe and the recovery opportunity is large.

2. Literature Review

Material flow cost accounting made internal losses financially legible by pricing waste as lost input rather than as a disposal item (Christ & Burritt, 2015). Circular economy research then moved the unit of analysis from the factory to the network, where value retention depends on actors who neither own nor control one another (Bocken et al., 2016; Kirchherr

et al., 2017). The accounting tools did not travel with the concept, so network losses stay unpriced.

Transparency research has grown quickly but treats visibility largely as an end state (Sodhi & Tang, 2019). Knowing where material goes is not the same as being able to act on that knowledge. Where disclosure is externally imposed, institutional theory predicts symbolic conformity rather than real change, since coercive pressure rewards the appearance of compliance (DiMaggio & Powell, 1983).

The capability and relational perspectives explain what closes the gap. Routines for sensing and reconfiguring resources are valuable because they are hard to copy (Barney, 1991; Teece, 2007), and relational rents come from governance and investments that partners build together (Dyer & Singh, 1998). Circular accounting capability fits this logic, since it is a joint routine rather than a purchased system.

Seven hypotheses follow.

- H1. Circular accounting capability increases material flow visibility.
- H2. Digital traceability increases visibility, more weakly than accounting capability.
- H3. Material flow visibility increases resource recovery action.
- H4. Circular governance increases inter-organisational coordination.
- H5. Coordination increases circular value creation.
- H6. Digital traceability increases circular value creation.
- H7. Digital traceability strengthens the coordination to value relationship.

3. Data and Methodology

Data come from a cross-sectional survey of agro-manufacturing enterprises spanning grain milling, oilseed processing, fruit and vegetable processing, dairy and sugar. One key informant per firm was targeted, holding operations, quality or sustainability responsibility. After screening for incomplete answers and insufficient informant knowledge, 520 usable responses were retained. Early and late respondents showed no meaningful differences on the focal constructs.

All focal constructs were measured reflectively on seven-point scales, using items adapted from prior work and refined through expert review and a pilot with practising managers. Resource recovery action was captured as realised behaviour, covering by-product rerouting, loss reduction projects and joint recovery investment, rather than intention. Circular governance covered contractual and relational mechanisms together. Firm size, age, export orientation and certification entered as controls.

Common method bias was assessed with a single-factor test, an unrelated marker variable and full collinearity variance inflation factors (Podsakoff et al., 2003). None of the diagnostics indicated a problem.

Partial least squares structural equation modelling suits a complex model that contains an interaction and is used for prediction as well as explanation (Hair et al., 2019). Paths were estimated with 10,000 bootstrap resamples and bias-corrected accelerated intervals, and the moderating effect through the two-stage approach (Becker et al., 2018). Discriminant validity used the heterotrait-monotrait ratio (Henseler et al., 2015), predictive power used PLSpredict (Shmueli et al., 2019), and endogeneity was checked with a Gaussian copula (Park & Gupta, 2012).

4. Results and Discussion

The measurement model performed well. Loadings exceeded 0.70, composite reliabilities ranged from 0.85 to 0.93, average variance extracted exceeded 0.50 for every construct, and all heterotrait-monotrait ratios stayed below 0.85. Inner model variance inflation factors remained under 3.3.

Circular accounting capability is the dominant driver of material flow visibility ($\beta = 0.532$, $p < 0.001$). Digital traceability also contributes, but its effect is roughly one third the size. Shared calculation, not shared connectivity, is what makes losses appear. A tracking system reports that material moved. An accounting capability states how much was lost, whose input it was and what it was worth.

The central result is a non-finding. Visibility does not translate into recovery action ($\beta = -0.052$, $p = 0.145$), and the confidence interval straddles zero, so H3 is rejected. Governance instead works through a separate route, raising coordination strongly ($\beta = 0.516$, $p < 0.001$), and coordination raises circular value creation ($\beta = 0.509$, $p < 0.001$). Seeing and doing run on different mechanisms.

Digital traceability matters as an amplifier rather than an initiator. Its direct effect on circular value creation is substantial ($\beta = 0.325$, $p < 0.001$) and it strengthens the coordination path through a small but reliable interaction ($\beta = 0.074$, $p < 0.05$). Simple slopes show that coordinated pairs extract more value when traceability is high, while high traceability with weak coordination gains little.

Together these results describe a decoupling of transparency from abatement. Disclosure regimes assume that information scarcity is the binding constraint. The evidence points instead to a capability and authority constraint, since data can be abundant while the right to act on a loss sits with a partner who bears none of its cost. That reading explains why traceability spending and recovery performance move independently.

5. Conclusion

Circular accounting capability, not digital traceability, is the primary source of material flow visibility in fragmented production networks. Visibility by itself changes nothing. Recovery action arrives through governance and coordination, with traceability strengthening the value that coordinated partners create. The circular transition is therefore a problem of shared calculation and decision rights rather than missing data.

Two limitations bound these claims. The design is cross-sectional, so the paths are associations supported by theory and an endogeneity check rather than demonstrated causation. The sample covers one sector with unusually high loss rates and seasonality, which aids identification but limits transfer to discrete manufacturing. Longitudinal and multi-sector work would test whether the decoupling persists as traceability systems mature.

6. Managerial and Social Implications

For managers the message is an order of operations. Investing in tracking technology before joint measurement and clear decision rights produces reporting rather than recovery. Buyers should build shared loss accounts with suppliers, agree how losses are allocated and priced, and only then scale the technology that carries the data. Traceability budgets pay off where coordination already exists.

For policymakers, mandating disclosure is necessary but insufficient. Rules that require firms to report material flows without settling who may act on a reported loss will generate compliance documents and little abatement. Instruments that fund joint measurement,

support cooperative recovery arrangements and reward verified reduction rather than verified reporting are more likely to shift outcomes.

The social stakes are concrete in agro-manufacturing. Unrecovered organic material is wasted food, avoidable emissions and lost income for small suppliers who absorb losses they cannot measure. Moving the calculative boundary so those losses become visible and assignable creates a basis for sharing recovery gains with the actors furthest upstream, linking net-zero goals to rural livelihoods.

Appendix

Table A1. Structural model results (10,000 bootstrap resamples)

Hypothesis	Path	β	p	Decision
H1	Accounting capability → Visibility	0.532	< 0.001	Supported
H2	Digital traceability → Visibility	0.177	< 0.01	Supported
H3	Visibility → Recovery action	-0.052	0.145	Not supported
H4	Governance → Coordination	0.516	< 0.001	Supported
H5	Coordination → Value creation	0.509	< 0.001	Supported
H6	Digital traceability → Value creation	0.325	< 0.001	Supported
H7	Traceability × Coordination → Value	0.074	< 0.05	Supported

Note. Standardised estimates. Firm size, age, export orientation and certification are controlled throughout.

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